

Live Translation Service — Implementation Plan

1. Goal

Build a browser-based live translation system for events.

The system allows an admin to create a program, add language streams, generate translator links, generate a public QR landing page, and let participants listen to any available translation stream from their phones.

The system will use:

- Cloudflare Realtime SFU for live audio distribution
 - Cloudflare Realtime TURN for connection fallback
 - Browser-based listener experience
 - Voice-only audio
 - No video
 - Participants are listen-only
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2. Core User Roles

Admin

Can:

- Log in with admin password
- Create/edit/delete programs
- Add language streams to a program
- Set translator password for each program
- Assign translators to language streams
- View stream status
- Generate QR code for program landing page
- Monitor active listeners per language

Translator

Can:

- Open translator URL: `/program_id/translate`
- Enter translator password
- Select assigned language stream
- Publish microphone audio

- See whether audio is live
- Mute/unmute
- Reconnect if needed

Participant / Listener

Can:

- Scan QR code
 - Open program page: `/program_id`
 - See available language streams
 - See which streams are currently live
 - Tap a language to start listening
 - Switch language streams
 - Listen only, never publish audio/video
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3. URL Structure

Public listener page

```
https://domain.com/{program_id}
```

Example:

```
https://translate.example.com/patna-event-2026
```

Translator page

```
https://domain.com/{program_id}/translate
```

Example:

```
https://translate.example.com/patna-event-2026/translate
```

Admin dashboard

```
https://domain.com/admin
```

4. Main Product Flows

4.1 Admin creates program

Admin enters:

- Program name
- Program ID / slug
- Event date
- Venue
- Expected audience size
- Admin notes
- Program status: draft / live / archived

System creates:

- Listener URL
- Translator URL
- QR code for listener URL

4.2 Admin adds language streams

For each language stream:

- Language name: Hindi, English, Tamil, Bengali, etc.
- Language code: hi, en, ta, bn
- Display order
- Active/inactive
- Assigned translator name
- Optional translator-specific password
- Cloudflare SFU room/session mapping

Example:

```
Program: patna-event-2026
```

```
Streams:
```

- ```
- Hindi
- English
- Bengali
- Tamil
```

Recommended model:

One Cloudflare SFU session per program  
One audio track per language stream  
Participants subscribe only to selected language track

Alternative model:

One Cloudflare SFU session per language  
Participants join only the selected language session

For simplicity and isolation, start with:

One SFU session per language stream

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## 5. Cloudflare Realtime Architecture

### 5.1 Basic architecture

Translator Browser  
↓ microphone audio  
Cloudflare Realtime SFU  
↓ audio track forwarding  
Participant Browser

TURN is used only when needed:

Translator / Participant  
↓ if direct WebRTC connection fails  
Cloudflare Realtime TURN  
↓  
Cloudflare Realtime SFU

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### 5.2 Audio-only configuration

Translator client:

- Request microphone only
- No camera permission

- Publish mono Opus audio
- Target bitrate: 32–48 kbps
- Echo cancellation: on
- Noise suppression: on
- Auto gain control: on
- Video: disabled

Listener client:

- No mic permission
  - No camera permission
  - Subscribe only
  - Receive one selected language audio track
  - Audio output through browser media element
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## 6. Suggested Tech Stack

### Frontend

- React / Next.js
- Tailwind or simple CSS
- WebRTC client integration with Cloudflare Realtime SFU
- QR code display/download
- Mobile-first listener page

### Backend

Can be one of:

- Cloudflare Workers
- Node.js API server
- Next.js API routes

Recommended for early version:

Cloudflare Pages + Cloudflare Workers + D1 / Postgres

### Database

Options:

- Cloudflare D1
- Supabase Postgres
- Neon Postgres

- Self-hosted Postgres

For MVP, use Postgres or D1.

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## 7. Data Model

### programs

```
id
slug
name
venue
event_date
status
admin_notes
created_at
updated_at
```

### language\_streams

```
id
program_id
language_name
language_code
display_order
is_active
is_live
current_track_id
cloudflare_session_id
created_at
updated_at
```

### translators

```
id
program_id
name
password_hash
allowed_stream_ids
created_at
updated_at
```

## listener\_sessions

```
id
program_id
language_stream_id
client_id
connected_at
disconnected_at
user_agent
network_info
```

## stream\_events

```
id
program_id
language_stream_id
event_type
timestamp
metadata
```

Example event types:

```
translator_connected
translator_disconnected
audio_started
audio_stopped
listener_joined
listener_left
connection_failed
reconnected
```

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## 8. Admin Dashboard

Admin dashboard features:

### Program list

- Program name
- Date
- Status
- Listener URL

- Translator URL
- QR code
- Total active listeners

## Program detail page

Shows:

- Program metadata
- Listener landing page URL
- Translator URL
- QR code
- Language streams
- Live/offline status per language
- Active listener count per language
- Translator connected/not connected
- Audio activity indicator
- Edit language streams
- Reset translator password

## QR code

Generate QR for:

```
https://domain.com/{program_id}
```

Admin should be able to:

- View QR
- Download PNG/SVG
- Print QR
- Copy listener URL

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## 9. Translator Page

URL:

```
{program_id}/translate
```

Flow:

1. Translator opens URL
2. Enters translator password
3. Selects assigned language stream



4. Grants microphone permission
5. Clicks “Go Live”
6. System publishes audio to Cloudflare SFU
7. Admin and listener pages show that language as “Live”

Translator UI:

- Program name
- Selected language
- Mic permission status
- Audio level meter
- Go Live / Stop
- Mute / Unmute
- Connection status
- Warning if audio input is silent
- Reconnect button

Important: Translator should use a stable internet connection and ideally a laptop with wired or high-quality Wi-Fi.

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## 10. Listener Page

URL:

`/ {program_id}`

Mobile-first layout:

Program Name

Choose your language

  

|           |            |
|-----------|------------|
| [ Hindi   | Live ]     |
| [ English | Live ]     |
| [ Bengali | Not Live ] |
| [ Tamil   | Live ]     |

  

Tap to listen

When listener taps a language:

- Browser starts audio
- Listener joins that stream as receive-only
- Player shows connected status
- Page shows selected language

- Option to switch language

Listener UI states:

### Language live

Hindi translation is live  
Tap to listen

### Language not live

Hindi translation has not started yet  
Waiting for translator...

### Connected

Listening to Hindi  
Audio live

### Reconnecting

Trying to reconnect...

### Disconnected

Connection lost  
Tap to reconnect

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## 11. Live Audio Indicator

Each language stream should show:

- Translator connected
- Audio track published
- Recent audio level detected
- Listener count

A stream should be marked “Live” only if:

```
translator_connected = true
AND audio_track_published = true
AND audio_activity_detected_recently = true
```

Suggested logic:

```
If audio level > threshold in last 5 seconds:
 show Live
Else:
 show Connected but silent
```

This avoids showing “Live” when the translator is connected but muted or silent.

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## 12. Security Model

### Admin

- Admin login required
- Store password as hash
- Rate-limit login attempts
- Use secure cookies/session
- Optional: allow only known admin email later

### Translator

- Translator URL is public but protected by password
- Password is per program or per translator
- Password should be hashed
- Translator session should expire after event
- Only translator clients can publish audio
- Participants must never receive publish permissions

### Participants

- No login required
  - Can only subscribe
  - Cannot publish audio/video
  - Cannot see admin controls
  - Cannot access translator controls
-

## 13. Permissions

Use server-generated short-lived tokens / session credentials.

### Admin token

Can:

- Create programs
- Edit streams
- View analytics

### Translator token

Can:

- Join assigned program
- Publish audio only
- Publish only to assigned language stream

### Listener token

Can:

- Join program
- Subscribe only
- No publishing

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## 14. Cost Estimate

For 5,000 listeners, 2 hours, audio-only:

Assume 50 kbps effective per listener:

```
50 kbps × 5,000 listeners = 250 Mbps
250 Mbps × 7,200 seconds = ~225 GB
```

Cloudflare Realtime SFU/TURN cost:

```
225 GB × $0.05/GB = $11.25
```

Since Cloudflare includes 1 TB/month free for Realtime SFU + TURN combined, the actual media cost for one such event may be zero if monthly usage stays under the free tier.

Other costs:

- Web hosting: likely free or very low
  - Database: free/low-cost tier initially
  - Domain: annual domain cost
  - Admin/backend compute: likely negligible for MVP
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## 15. Mobile Browser Considerations

Participants will use mobile browsers.

Important browser constraints:

- Audio must start after a user tap
- Do not rely on autoplay
- Show a clear “Tap to Listen” button
- Keep reconnect button visible
- Test iPhone Safari and Android Chrome
- Test Bluetooth earphones
- Test lock screen behavior
- Test low-power mode
- Test mobile data and venue Wi-Fi

Because WebRTC background audio behavior can vary by browser/OS, event instructions should say:

For best experience, keep this page open after starting audio.  
If audio stops, reopen the page and tap Listen again.

Before the real event, run a field test with real iPhones and Android phones.

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## 16. MVP Scope

Build first version with:

- Admin login
- Create program
- Add language streams
- Generate QR code
- Translator login
- Translator publishes one language audio stream
- Listener page lists languages
- Listener subscribes to one language
- Live/silent/offline indicators

- Basic listener counts
- Basic logs

Avoid in MVP:

- Payments
  - User accounts for listeners
  - Native apps
  - Chat
  - Video
  - Recording
  - Complex analytics
  - Multi-admin roles
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## 17. Testing Plan

### Functional testing

- Create program
- Add languages
- Generate QR
- Translator login works
- Wrong password blocked
- Translator can publish audio
- Listener can join
- Listener can switch language
- Listener cannot publish
- Admin sees live status

### Device testing

Test on:

- Android Chrome
- iPhone Safari
- iPhone Chrome
- Low-end Android phone
- Bluetooth earphones
- Wired earphones
- Phone speaker
- Lock screen
- Poor network

## Load testing

Before event:

- Simulate 500 listeners
- Simulate 1,000 listeners
- Simulate 5,000 listeners if possible
- Track Cloudflare usage
- Track join success rate
- Track average connection time
- Track audio dropouts
- Track reconnects

## Venue testing

At venue:

- Test translator upload network
  - Test participant mobile data
  - Test venue Wi-Fi if available
  - Test QR scanning from physical banner
  - Test joining from the back of the venue
  - Test backup translator device
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## 18. Operational Checklist for Event Day

Before event:

- Create program
- Add language streams
- Generate QR code
- Print QR signage
- Test translator links
- Test each language stream
- Keep backup translator device ready
- Keep backup internet hotspot ready
- Keep admin dashboard open

During event:

- Monitor translator connected status
- Monitor audio activity
- Monitor listener counts
- Monitor stream silence warnings
- Keep support person available for participant issues

After event:

- Archive program
  - Export basic analytics
  - Review dropouts and issues
  - Reset translator passwords
- 

## 19. Recommended MVP Build Sequence

### Phase 1: Admin + database

- Admin login
- Program CRUD
- Language stream CRUD
- QR generation

### Phase 2: Translator flow

- Translator login
- Mic permission
- Publish audio stream
- Audio level meter
- Live/offline state

### Phase 3: Listener flow

- Program landing page
- Language list
- Subscribe to selected stream
- Switch stream
- Reconnect handling

### Phase 4: Monitoring

- Active listener count
- Translator status
- Audio activity indicator
- Error logs

### Phase 5: Field readiness

- Mobile testing
  - Load testing
  - Venue testing
  - Event-day dashboard
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## 20. Key Product Principle

This is not a meeting app.

It is:

One translator publishes.  
Thousands of people listen.

Therefore, keep the system optimized for:

- voice only
- receive-only participants
- fast joining
- clear language selection
- reliable reconnection
- low operational complexity